Ayoub Sassi

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Summary

CS Master's (GPA 3.96) with published IEEE research in deep learning and hands-on experience building RL, computer-vision, and RAG/LLM systems. Seeking ML/AI Engineer roles.

Experience

Research Assistant **Mar. 2024 – Present**
University of Michigan-Flint *Flint, MI*

- Designed and trained deep learning models to optimize linear precoding in multi-user VLC systems, achieving near-optimal interference mitigation with **225× lower computational time** compared to the baseline method.
- Research paper published at **IEEE Open Journal of the Communications Society (OJCOMS).**
- Built and trained an RL agent integrated with Bird's Eye View fusion in the CARLA simulator to enable safe and efficient autonomous intersection navigation, achieving **92% collision avoidance.**
- Research paper published and presented at **IEEE International Symposium on Circuits and Systems (ISCAS 2025)**, London, UK.

Summer Intern **Jun. 2023 – Aug. 2023**
Pixmantic *Ariana, Tunisia*

- Designed and implemented a RAG-based medical chatbot using LangChain, FAISS, and Streamlit, integrating LLMs and document embeddings to retrieve and generate context-aware medical responses.

Education

University of Michigan Flint **Sept. 2024 – Apr. 2026**
MS in Computer Science, GPA: 3.96/4.00 *Flint, MI*

Higher School of Communication of Tunis (SUP'COM) **Sept. 2021 – Jul. 2024**
National Engineering diploma, ICT Engineering *Tunis, Tunisia*

Preparatory School of Engineering Studies of Tunis **Sept. 2019 – June 2021**
Ranked 42/1200 in the national exam *Tunis, Tunisia*

Projects

Voice AI QA Bot **Jun. 2026 – Jul. 2026**

∗ Built an automated voice-testing pipeline that places real outbound calls to a company's AI phone receptionist via Twilio and bridges live call audio to Google's Gemini Live API in real time, simulating natural patient conversations across 10 test scenarios; analyzed 11 recorded calls to identify and document 5 bugs, including a systemic identity-verification failure affecting 82% of calls.

World Cup Storyteller **Jan. 2026 – Apr. 2026**

∗ Comparative NLP app implementing Zero-Shot, Few-Shot, and RAG pipelines for sports narrative generation; evaluated with LLM-as-a-Judge across 3 independent models; deployed on Hugging Face Spaces.

Road Signs Detection **Sept. 2024 – Dec. 2024**

∗ Developed a YOLO-based road sign detection model for autonomous driving applications, achieving 96.7% precision and 89.6% recall on 801 test images across 14 traffic sign and signal classes.

Technical Skills

Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision; RAG, LLMs, LangChain, Llamaindex, Vector Databases (FAISS, Qdrant), Prompt Engineering, Agentic AI; Python, SQL; PyTorch, TensorFlow, Scikit-learn, Pandas; Git, Streamlit, CARLA, Kaggle, Google Colab

Certificates

∗ AWS Certified Cloud Practitioner
∗ Supervised Machine Learning: Regression and Classification
∗ Advanced Learning Algorithms